Real Effects of Inflation Expectation Uncertainty

Nam T. Vu*

Abstract

Using survey data on consumer expectations, we find that a rise in inflation expectation uncertainty lowers both current and expected consumption spending. The relationship holds after controlling for income, financial conditions, demographics, the level of expected inflation, and five widely used measures of aggregate uncertainty. To rationalize this result, we incorporate uncertainty about how the household cognitively discounts its future inflation expectations into an otherwise standard New-Keynesian model. Unlike in previous work, the degree of cognitive discounting is itself time-varying and subject to an uncertainty shock that widens its distribution. Estimating the model by simulated method of moments yields a steady-state cognitive discount factor of 0.752, implying an information half-life of roughly two and a half quarters. Combined with precautionary saving motives, the increases in markup and expected real interest rate arising from inflation expectation distortion amplify the fall in consumption.

Keywords: Inflation expectations, consumption response, cognitive discounting, stochastic volatility

JEL Classifications: E31, D12

*Corresponding author. Miami University, Department of Economics: 800 E. High St., Oxford, OH, 45056, the United States of America. Email: vunt@miamioh.edu.

1. Introduction

Between 2020 and 2022, as U.S. inflation rose to its highest level in four decades, consumers faced both rising prices and growing disagreement over where prices were headed, following COVID-19 and the resulting supply chain disruptions. In this environment, uncertainty about future inflation can rise even among consumers who share similar expected inflation rates. This paper asks whether such uncertainty, distinct from the level of expected inflation itself, has independent effects on household consumption. Understanding the real effects of dispersion in inflation expectations is relevant for consumers and policymakers alike.

Consumer perceptions play a crucial role in transmitting unexpected shocks to the economy and the policy that follows. The Great Recession of 2008, the European debt crisis, and the more recent COVID-19 pandemic have elevated the need for a better understanding of the impact of consumer expectations and, perhaps more importantly, the channels through which they impact vital macroeconomic variables. These events have rekindled an extensive literature focusing on the effects of expectations (e.g., Branch and McGough, 2009, Coibion and Gorodnichenko, 2015, and De Grauwe and Macchiarelli, 2015) and precisely how inflation expectations impact key macroeconomic aggregates (e.g., Coibion et al., 2020). Another related literature focuses on the overall implications of uncertainty on macroeconomic aggregates (e.g., Jurado et al., 2015 and Ludvigson et al., 2021). At the intersection of these strands of the literature, this paper investigates the real effects of uncertainty shocks on consumers' inflation expectations.

To that end, we first examine how consumption is empirically related to inflation expectation uncertainty. Based on the Federal Reserve Bank of New York survey data, our benchmark measure of inflation expectation uncertainty is the standard deviation of each respondent's expected inflation. Our regression results show that an increase in such uncertainty is associated with a fall in the respondent's real and expected consumption spending, controlling for their income, financial condition, state of residence, and other demographic characteristics.[1]

[1] We use the term *real* for consumption throughout the paper to refer to *actual* consumption, as opposed to *expected* consumption, which is not yet realized in the current period.

In addition, the robustness of the regression results when controlling for movements of aggregate inflation suggests that the resulting change in consumption is not necessarily due to an increase in aggregate inflation.

Given that uncertainty measures are known to be strongly correlated, we next examine whether inflation expectation uncertainty merely reflects information about other uncertainty sources, controlling for aggregate uncertainty arising from financial markets, commodities, economic policies, and overall macroeconomic conditions. We find our baseline regression results robust to these sources. In other words, the effects of inflation expectation uncertainty on consumption are not purely driven by these uncertainty channels.

The robustness of our empirical results motivates us to consider an economic environment in which agents exhibit uncertainty about next-period inflation. Specifically, we study a dynamic model in which the household cognitively discounts expected inflation, subject to exogenous level and stochastic volatility shocks. An unexpected change in the stochastic volatility component of this cognitive discount factor is the inflation expectation uncertainty shock in the model. This shock distorts the household's discounting of the future by widening the distribution of its expected inflation. Our decision to allow the household to discount expected inflation is motivated by Gabaix (2020), who proposes a novel cognitive discounting factor to capture how poorly agents perceive the state of the economy.

We opt for cognitive discounting over alternative approaches to bounded rationality for two reasons. First, models of sticky information (Mankiw and Reis, 2002) and noisy signals (Sims, 2003) generate heterogeneity in expectations but do not naturally produce time-varying uncertainty in how agents *discount* future inflation. In contrast, introducing stochastic volatility into the cognitive discount factor (m_t in the model) provides a channel through which the dispersion of inflation perceptions can vary over time. This addition allows us to map to the inflation expectation uncertainty we measure in the data.[2] Second, cognitive discounting nests the standard rational expectations benchmark as a special case (when $m_t = 1$), which allows us to isolate the contribution of inflation expectation uncertainty without al-

[2]Woodford (2019) considers a model in which decision makers face a finite time horizon and highlights the importance of incorporating such a horizon in the optimality conditions.

tering other features of the model. The resulting environment requires only one additional state variable, the stochastic volatility component, to generate the uncertainty channel documented in the empirical analysis.

The rest of the economic environment consists of households working to finance their consumption and housing expenditures, in the spirit of Iacoviello (2005). An entrepreneur owns a firm that uses labour and commercial real estate to produce goods, and pledges their commercial real estate holdings as collateral for borrowing funds. A continuum of retailers distributes goods in a monopolistically competitive market. We select model parameters based on external targets and estimate the remaining parameters using a simulated method-of-moments. Our estimation yields a steady-state cognitive discount factor of 0.752, implying that the information the agents receive today has an average half-life of around two and a half quarters.

We next examine the model's responses to an increase in inflation expectation uncertainty and find they are qualitatively consistent with the empirical evidence. The model predicts that a positive one-standard-deviation shock to inflation expectation uncertainty leads to a decrease in consumption on impact and multiple periods afterward. This decline accompanies the falls in housing, hours, output, and inflation, and the rise in markup. This qualitative consistency in consumption responses across the empirical analysis and the model provides a potential mechanism for understanding the empirical result.

Intuitively, a rise in the mean-preserving spread of the cognitive discount factor increases expected consumption growth, which dampens current-period consumption due to precautionary saving motives. Two channels amplify such a contraction. First, a surge in markup induces firms to produce less output, requiring fewer labour hours and less commercial real estate. Second, an increase in the stochastic volatility of the cognitive discount factor distorts the household's inflation expectations, raising the expected real interest rate and prompting the household to save more.

Our paper's contribution is twofold. First, we empirically document that consumption responds negatively to an increase in inflation expectation uncertainty, and that this result holds even after controlling for income and various financial and demographic characteristics. Second, we provide a potential mechanism to understand the empirical results via an otherwise standard New-Keynesian model in which both the household's cognitive discounting of future inflation expectations and the uncertainty concerning such discounting are time-varying. We estimate the model and find that it can qualitatively explain the decline in consumption documented in the data.

Our paper is related to four main strands of the literature. The first strand studies the implications of consumer expectations on macroeconomic aggregates, showing that consumer expectation plays a vital role in the household's savings and consumption and that inflation expectation depends significantly on the consumers' previous experiences.[3] For instance, Malmendier and Nagel (2016) study how individuals form expectations about future inflation under an adaptive learning framework and find experiences strongly predict differences in expectations. Bachmann et al. (2015) use the microdata from the Michigan Surveys of Consumers and show that a one-percentage-point increase in expected inflation reduces households' probability of having a positive attitude towards spending by about 0.5 percentage points in low-interest-rate environments. Using a large multi-country survey, Duca-Radu et al. (2021) find positive, economically relevant spending changes in response to consumers' beliefs about future inflation, often amplified during low-interest-rate episodes. Using Michigan Survey data, Binder (2017) finds that inflation uncertainty reduces consumers' reported readiness to buy durable goods. Fuest and Schmidt (2026) also study inflation expectation uncertainty within a New-Keynesian framework and find it negatively affects inflation and the output gap, with the demand channel outweighing the supply channel. However, the channels they consider differ from the cognitive-discounting channel we study. More recently, Hilscher et al. (2026) use long-dated inflation swap contracts to construct a new measure of expected inflation and highlight the importance of fully accounting for inflation risk.

Second, our paper complements a strand of the literature on how bounded rationality

[3]See, e.g., Branch (2004), Branch and McGough (2009), Souleles (2004), Lanne et al. (2009), Pfajfar and Santoro (2010), Armantier et al. (2013), Easaw et al. (2013), Bachmann et al. (2015), Madeira and Zafar (2015), Malmendier and Nagel (2016), Vellekoop and Wiederholt (2019), Dräger et al. (2020), and more recently, D'Acunto et al. (2021), and Duca-Radu et al. (2021), among others.

and distortions in consumer expectations affect macroeconomic aggregates, a literature that finds that incorporating such distortions is essential for macroeconomic models to be consistent with the data.[4] For example, Reis (2006) studies a theoretical framework in which consumers rationally update their information and re-compute their optimal consumption plans sporadically and finds that individual consumption is sensitive to unexpected news but not predictable events. Branch (2007) shows that a sticky information model with a time-varying distribution structure is consistent with the Michigan inflation expectation survey. Angeletos and Lian (2018) connect imperfect information to muted forward guidance effects, which is an example of how agents discount future expectations. Gabaix (2020) considers a widely used New-Keynesian model and introduces a novel cognitive discounting parameter that quantifies how poorly agents understand future economic disturbances.[5]

Third, our paper is also related to a broad literature that studies the overall implication of uncertainty on macroeconomic aggregates both theoretically (e.g., Bloom, 2009, Fernández-Villaverde et al., 2011, Bi et al., 2013, and Fernández-Villaverde et al., 2015, among others) and empirically (e.g., Jurado et al., 2015, Vu and Wu, 2020, Ludvigson et al., 2021, Berg and Vu, 2021, Coibion et al., 2024, Coibion and Gorodnichenko, 2026, and Candia et al., 2026, among others).

Fourth, our paper complements another strand of the literature studying the overall interplay between inflation and other macroeconomic aggregates (see, e.g., Mallick and Mohsin, 2010, Galati et al., 2023, and Kawamoto et al., 2025). Here, we contribute by showing that uncertainty with respect to inflation expectation does have an impact on consumption, distinct from the level effect of consumption.

The paper proceeds as follows. Section 2 presents the empirical evidence that inflation expectation uncertainty is negatively related to consumption. Building on this result, Section 3 presents a New-Keynesian model where the consumer discounts future inflation expecta-

[4]See, e.g., Mankiw and Reis (2002), Reis (2006), Branch (2007), Branch and Evans (2011a), Branch and Evans (2011b), Adam and Marcet (2011), Massaro (2013), Di Bartolomeo et al. (2016), Dräger (2016), Pecora and Spelta (2017), Gasteiger (2018), Gelain et al. (2019), Hommes and Lustenhouwer (2019), Jump and Levine (2019), Carroll et al. (2020), and more recently Gabaix (2020), among others.

[5]Farhi and Werning (2019) extend Gabaix's bounded rationality framework to incomplete markets settings.

tions. Section 4 presents the model calibration and estimation to the data. Section 5 discusses the model's key implication that consumption responds negatively to increased inflation expectation uncertainty, including the role of financial frictions. Section 6 documents the extent to which the model's predictions align with the survey evidence and with aggregate consumption data. The paper concludes in Section 7.

2. Empirical Evidence

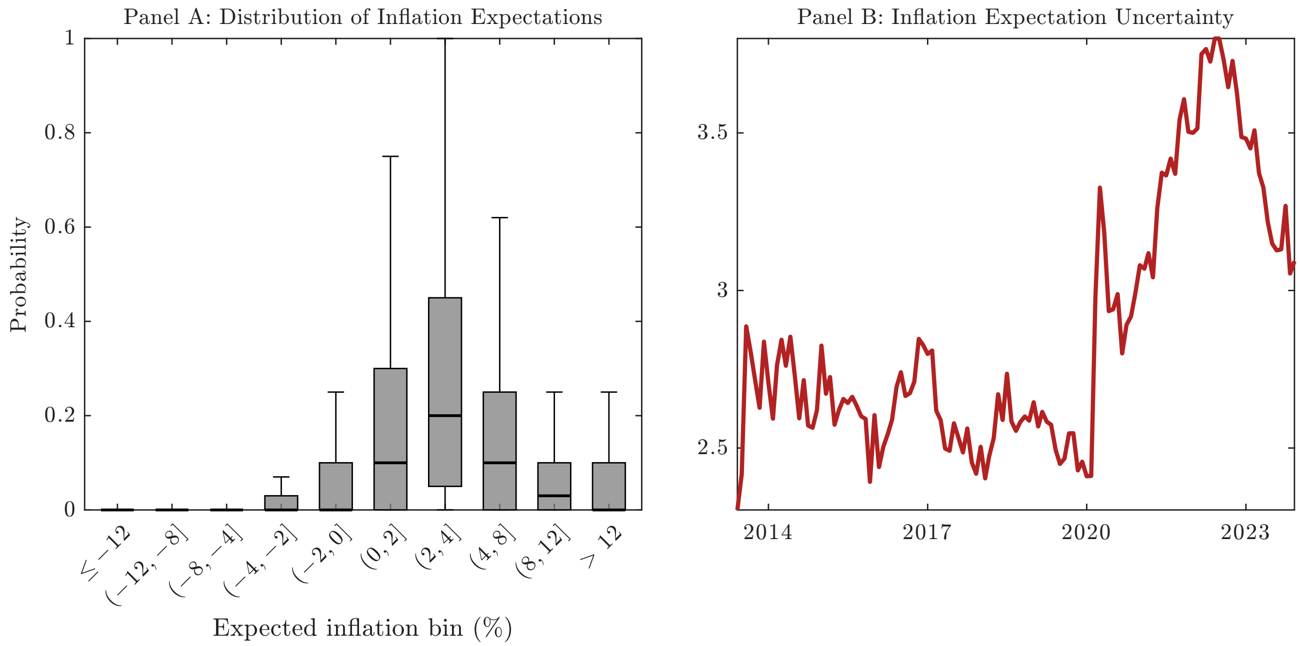
In this section, we examine how consumption is related to increased uncertainty about inflation expectations. We describe the dataset, followed by the empirical specifications, their results, and a series of sensitivity analyses. Our main result is that both real and expected consumption are negatively related to inflation expectation uncertainty.

2.1. Data Sources and Variable Construction

The data used in the empirical analysis are from the Federal Reserve Bank of New York's Survey of Consumer Expectations, in which respondents report their likelihood that inflation will be in a particular range. The raw data are monthly and span from 2013:M6 to 2023:M9. Using the SCE allows us to explore the empirical distributions of consumers' expectations, not restricted to the first moment, and, more importantly, to link these responses to consumption spending and other individual characteristics.

Our benchmark uncertainty measure is the standard deviation of each respondent's inflation expectation, where inflation expectation is based on the question "In your view, what would you say is the percent chance that, over the next 12 months...," in which the survey participants are asked to provide the probability that the rate of inflation will take on different values. Fig. 1 plots the distribution of inflation expectations across all respondents and a time-series plot of our benchmark uncertainty measure. Panel A is a box-and-whisker plot of respondents' inflation expectations across 10 bins, ranging from "-12% or less" to "12% or more," pooling all periods (i.e., from 2013:M6 to 2023:M9). Panel A shows that, across all

Figure 1: Descriptive Statistics: Inflation Expectation Uncertainty



Notes: Panel A plots the distribution of inflation expectation in the next 12 months *pooling* all periods in the sample (2013:M6 to 2023:M9). Here, inflation expectation is based on the probability that the inflation rate will take on different values. Panel B plots the evolution of the average inflation expectation uncertainty across all survey respondents over time. This uncertainty measure is the standard deviation of each respondent's inflation expectation, where inflation expectation is based on the same question used in Panel A. Source: Author's calculations.

periods in the sample, survey respondents mostly expect inflation to be between 0% and 8%. Panel B plots the average benchmark uncertainty over the entire sample, showing a significant increase in average inflation expectation uncertainty post-COVID-19.

We consider two variables that capture the consumption response: the actual reported (real) consumption and expected consumption. Real consumption is a binary variable that captures whether the survey respondent has recently made a large purchase. Specifically, the variable is constructed based on the question "And thinking about the more recent past, did any members of your household (including you) make any of the following large purchases during the last 4 months?", of which the answer can be one or more of the following: home appliances, electronics, computers or cell phones, furniture, home repairs, improvements or renovations, car or other vehicles, or trips and vacations. Expected consumption (change) is a continuous variable that answers the question, "By about what percent do you expect your total household spending to increase/decrease [over the next 12 months]? Please give your

Table 1: Real Consumption Expenditure and Inflation Expectation Uncertainty

	Dependent Variable: Real Consumption				
	(1)	(2)	(3)	(4)	(5)
Inflation Expectation Uncertainty	-0.0196***	-0.0364***	-0.0364***	-0.0359***	-0.0358***
	(0.00260)	(0.00233)	(0.00233)	(0.00234)	(0.00233)
Expected Inflation (Level)	0.00486***	-0.00148	-0.00146	-0.000212	-0.000307
	(0.00125)	(0.00161)	(0.00162)	(0.00161)	(0.00166)
Better Credit			0.00915	-0.000353	-0.0000328
			(0.0196)	(0.0205)	(0.0206)
Better Finance				0.134***	0.138***
				(0.0239)	(0.0237)
Better Expected Finance					-0.0118
					(0.0303)
Constant	-0.455***	-0.478***	-0.480***	-0.550***	-0.544***
	(0.0102)	(0.0536)	(0.0540)	(0.0563)	(0.0647)
Controls	-	✓	✓	✓	✓
R-Squared	0.00154	0.205	0.205	0.206	0.206
χ^2	64.07	6208.2	6661.1	6436.3	7572.1
Observations	71396	34876	34876	34876	34876

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Controls included in the table: homeownership, female, race (whether the respondent is Caucasian), employment status, location (state), year, quarter, expected change in income, and current income indicators (less than $30K, from $30K to $75K, from $75K to $200K, or $200K or above). The dependent variable is real consumption, a binary indicator of whether the survey respondent has recently made a large purchase. Source: Author's calculations.

best guess."

To examine the effects of inflation expectations on consumption, we utilize several sets of control variables in the regression analysis. The first set - *Better Finance*, *Better Credit*, and *Better Expected Finance* - reflects the respondent's current financial situation and their expectations for their future financial situation. The second set - *Employed*, *Home Owner*, *Caucasian*, *Female* - contains survey respondents' employment status and other demographic characteristics. The third set includes the respondent's annual income, classified into four bins: less than or equal to $29,999, $30,000 to $74,999, $75,000 to $199,999, and $200,000 or more. For brevity, Appendix A provides a detailed description of these variables and the corresponding survey questions.[6]

[6]Table A.1 in the Appendix presents the means and standard deviations of these variables. Fig. A.1 in the Appendix plots the distribution of inflation expectations by employment, marital, and homeownership status. On average, married, employed homeowners tend to have lower inflation expectations.

2.2. *Empirical Results*

2.2.1. *Real Consumption Expenditure*

To examine how inflation expectation uncertainty is related to *real* consumption spending, we consider the following Probit regression

$$Y_{it} = \alpha + \beta_1 \mathrm{IEU}_{it} + \mathbb{B}\mathbb{X}_{it} + c_i + d_t + \varepsilon_{it}, \tag{1}$$

where Y_{it} is a binary variable indicating whether a survey respondent i has recently made a large purchase, examples of which are home appliances, electronics, computers or cell phones.[7] IEU_{it} is inflation expectation uncertainty, c_i contains the individual-specific effects (captured by demographic controls), d_t contains time fixed effects, and $\mathbb{X}_{it}$ contains the controls that reflect respondent i's expectations of their financial situation, level of income, the expected change in income, employment status, and other demographic characteristics. We also control for each individual respondent's average inflation expectation in Eqn. (1). The related literature motivates our controlling for the respondent's financial and demographic characteristics (Malmendier and Nagel, 2016; Souleles, 2004). Given that the dependent variable Y_{it} is binary, we estimate Eqn. (1) using a Probit regression with time fixed effects and robust standard errors.[8]

Table 1 presents the estimation results for Eqn. (1). Across all specifications, the estimated coefficients for IEU_{it} are negative and statistically significant. This result shows that inflation expectation uncertainty is negatively associated with the probability that a survey respon-

[7] Specifically, the variable is constructed based on the question "And thinking about the more recent past, did any members of your household (including you) make any of the following large purchases during the last 4 months?", of which the answer can be one or more of the following: home appliances, electronics, computers or cell phones, furniture, home repairs, improvements or renovations, car or other vehicles, or trips and vacations. Our choice to use durable consumption in the benchmark specification is motivated by recent work highlighting the importance of durable good consumption for understanding key macroeconomic dynamics (McKay and Wieland, 2021, 2022; Kim et al., 2024).

[8] To address potential issues in terms of unobserved heterogeneity vis-à-vis the use of fixed effects, we have also considered alternative specifications to our benchmark probit model with fixed effects. Our results are generally consistent when a random-effects probit or logit model is used in lieu of the benchmark probit regression with fixed effects. The results of these estimations are presented in Table A.4 in the Appendix (Columns 2 and 3).

Table 2: Real Consumption Expenditure - Other Sources of Uncertainty

	Dependent Variable: Real Consumption					
	(1)	(2)	(3)	(4)	(5)	(6)
	VIX	Oil	Gold	EPU	JLN-3	JLN-12
Inflation Expectation Uncertainty	-0.0360***	-0.0359***	-0.0358***	-0.0363***	-0.0358***	-0.0358***
	(0.00235)	(0.00234)	(0.00232)	(0.00237)	(0.00231)	(0.00231)
Expected Inflation (Level)	-0.000120	-0.000229	-0.000407	-0.0000341	-0.000395	-0.000435
	(0.00165)	(0.00166)	(0.00166)	(0.00166)	(0.00166)	(0.00165)
Better Credit	-0.00183	-0.00164	0.00118	-0.00548	0.000996	0.00116
	(0.0205)	(0.0207)	(0.0207)	(0.0205)	(0.0207)	(0.0208)
Better Finance	0.142***	0.140***	0.136***	0.145***	0.137***	0.136***
	(0.0237)	(0.0239)	(0.0238)	(0.0239)	(0.0238)	(0.0238)
Better Expected Finance	-0.0111	-0.0106	-0.0130	-0.00728	-0.0127	-0.0128
	(0.0304)	(0.0303)	(0.0302)	(0.0304)	(0.0303)	(0.0303)
Aggregate Uncertainty	0.0129***	0.00120***	-0.0163***	0.00181***	-0.494***	-1.239***
	(0.00241)	(0.000275)	(0.00324)	(0.000137)	(0.149)	(0.309)
Constant	-0.895***	-0.601***	-0.234***	-0.770***	-0.0762	0.714**
	(0.0708)	(0.0639)	(0.0859)	(0.0631)	(0.137)	(0.300)
Controls	✓	✓	✓	✓	✓	✓
R-Squared	0.206	0.206	0.206	0.208	0.206	0.206
χ^2	8715.3	8698.2	7791.3	10244.3	8762.2	8549.0
Observations	34876	34876	34876	34876	34876	34876

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Controls included in the table: homeownership, female, race (whether the respondent is Caucasian), employment status, location (state), year, quarter, expected change in income, and current income indicators (less than $30K, from $30K to $75K, from $75K to $200K, or $200K or above). The dependent variable is real consumption, a binary indicator of whether the survey respondent has recently made a large purchase. Source: Author's calculations.

dent recently made a large purchase. In other words, an increase in inflation expectation uncertainty is associated with a negative real consumption response. Notably, while current personal finance shows economic and statistical significance in Table 1, these effects do not qualitatively alter the negative relationship between IEU_{it} and consumption spending.

Given the well-documented correlation between measures of uncertainty, we examine the extent to which IEU_{it} reflects information regarding other sources of uncertainty. To that end, we consider five well-known measures of uncertainty commonly used in the related literature (the last of which we include at two horizons, yielding six specifications) and include them as controls in the baseline specification in Table 1. The first measure is the Chicago Board Options Exchange's (CBOE) Volatility Index (*VIX*), which captures the stock market's expectation of volatility. The second and third measures are the CBOE Crude Oil ETF Volatility Index (*Oil*) and the CBOE Gold Volatility Index (*Gold*), which capture uncertainty arising

Table 3: Expected Consumption Expenditure and Inflation Expectation Uncertainty

	Dependent Variable: Expected Consumption				
	(1)	(2)	(3)	(4)	(5)
Inflation Expectation Uncertainty	0.125*	0.0671	0.0676	0.0664	0.0623
	(0.0722)	(0.0846)	(0.0845)	(0.0843)	(0.0836)
Inflation Expectation Uncertainty (1-lag)	-0.291***	-0.190***	-0.190***	-0.190***	-0.192***
	(0.0572)	(0.0612)	(0.0612)	(0.0613)	(0.0613)
Inflation Expectation Uncertainty (2-lag)	0.0974	-0.0734	-0.0736	-0.0738	-0.0756
	(0.0600)	(0.0507)	(0.0507)	(0.0507)	(0.0507)
Inflation Expectation Uncertainty (3-lag)	-0.0431	-0.140***	-0.139***	-0.139***	-0.139***
	(0.0772)	(0.0534)	(0.0533)	(0.0533)	(0.0533)
Dependent Variable (1-lag)	-0.230**	-0.591***	-0.591***	-0.591***	-0.592***
	(0.114)	(0.0457)	(0.0457)	(0.0457)	(0.0456)
Dependent Variable (2-lag)	-0.134**	-0.409***	-0.409***	-0.409***	-0.409***
	(0.0652)	(0.0709)	(0.0709)	(0.0709)	(0.0709)
Dependent Variable (3-lag)	-0.0195	-0.228***	-0.228***	-0.228***	-0.228***
	(0.0165)	(0.0397)	(0.0397)	(0.0397)	(0.0398)
Expected Inflation (Level)	0.0811***	0.0937***	0.0933***	0.0926***	0.0905***
	(0.0290)	(0.0315)	(0.0314)	(0.0315)	(0.0317)
Better Credit			-0.242	-0.235	-0.226
			(0.370)	(0.370)	(0.370)
Better Finance				-0.461	-0.312
				(0.326)	(0.317)
Better Expected Finance					-0.912***
					(0.342)
Constant	-0.305	-1.134	-1.092	-0.746	-0.0771
	(0.297)	(0.853)	(0.855)	(0.835)	(0.823)
Controls	-	✓	✓	✓	✓
Joint IEU Effects	5.043	19.69	19.69	19.69	19.89
R-Squared	0.107	0.290	0.290	0.290	0.290
χ^2/F	6.962	38.01	36.39	34.30	32.50
Observations	65129	41977	41977	41977	41977

Standard errors in parentheses
$^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$

Notes: Controls included in the table: homeownership, female, race (whether the respondent is Caucasian), employment status, location (state), expected change in income, and current income indicators (less than \$30K, from \$30K to \$75K, from \$75K to \$200K, or \$200K or above). The dependent variable is the expected change (in percentage) in expected consumers' expenditure. *Joint IEU Effects* denotes a joint test on the null hypothesis that the sum of the lag effects of inflation expectation uncertainty equals zero. Source: Author's calculations.

from oil and gold, respectively. The fourth measure is the Baker et al. (2016) Economic Policy Uncertainty Index for the U.S. (*EPU*). The fifth measure is the Jurado et al. (2015) macroeconomic uncertainty index at the three-month (*JLN-3*) and twelve-month (*JLN-12*) horizons, which captures uncertainty regarding the state of the macroeconomy. We present our results in Table 2, where the aggregate uncertainty measures in Columns (1)-(6) are *VIX, Oil, Gold, EPU, JLN-3,* and *JLN-12*, respectively.

The results controlling for these aggregate uncertainty measures in Table 2 remain consistent with the baseline results vis-à-vis the effects of IEU_{it} on consumption in Table 1. Such consistency is not surprising, considering that across individual respondents of the survey, IEU_{it} exhibits little to no correlation with the commonly used measures of aggregate uncertainty mentioned above on average: 0.023 for the *VIX*, 0.028 for *Oil*, 0.051 for *Gold*, 0.025 for *EPU*, and around 0.004 for the two Jurado et al. (2015) measures. Overall, the results in Table 2 suggest that the effects of IEU_{it} on real consumption are not driven by other channels of uncertainty that the literature has carefully documented.

2.2.2. Expected Consumption Expenditure

To examine how inflation expectation uncertainty is related to *expected* consumption spending, we consider the following fixed-effects panel regression

$$\Delta Y_{it} = \alpha + \sum_{s=1}^{3} \rho_s \Delta Y_{it-s} + \sum_{s=0}^{3} \beta_s \mathrm{IEU}_{it-s} + \mathbb{B}\mathbb{X}_{it} + c_i + \varepsilon_{it}, \tag{2}$$

in which ΔY_{it} is the percentage change of consumer i's expected expenditure in month t, IEU_{it} is inflation expectation uncertainty, c_i contains the individual-specific effects, and $\mathbb{X}_{it}$ contains the set of controls identical to the one used in Eqn. (1).

Table 3 presents the estimation results for Eqn. (2). Across all specifications, the cross-sectional unit is the individual survey respondent, and the time unit is the month.[9] The estimated coefficients (first-lag) for IEU_{it} are negative and statistically significant across all specifications. We also find that the total effects of all IEU_{it} lags are significantly different from zero, the test statistics of which are presented at the bottom of Table 3. These results suggest that respondents with higher inflation expectation uncertainty are more likely to decrease their *expected* consumption spending. These effects are significant after controlling for the respondent's financial and economic conditions and other characteristics.[10] We also note that the significance of inflation expectation uncertainty with a lag is consistent with the

[9]Our results are consistent when IEU_{it} is consumer i's inflation expectations' interquartile range in month t. The cross-sectional average of this alternative measure is plotted in Fig. A.2 in the Appendix.

[10]Even though we consider three lags of the dependent variable in Eqn. (2), our results are qualitatively consistent when other lag structures are considered (e.g., four and six).

Table 4: Expected Consumption - Controlling for other Sources of Uncertainty

	Dependent Variable: Expected Consumption					
	(1)	(2)	(3)	(4)	(5)	(6)
	VIX	Oil	Gold	EPU	JLN-3	JLN-12
Inflation Expectation Uncertainty	0.0768	0.0774	0.0761	0.0767	0.0756	0.0757
	(0.0876)	(0.0877)	(0.0875)	(0.0878)	(0.0875)	(0.0875)
Inflation Expectation Uncertainty (1-lag)	-0.191***	-0.191***	-0.191***	-0.191***	-0.191***	-0.191***
	(0.0625)	(0.0625)	(0.0624)	(0.0626)	(0.0625)	(0.0624)
Inflation Expectation Uncertainty (2-lag)	-0.0824	-0.0832	-0.0825	-0.0819	-0.0822	-0.0826
	(0.0516)	(0.0516)	(0.0516)	(0.0516)	(0.0516)	(0.0516)
Inflation Expectation Uncertainty (3-lag)	-0.106*	-0.107*	-0.105*	-0.106*	-0.106*	-0.107*
	(0.0553)	(0.0554)	(0.0553)	(0.0554)	(0.0554)	(0.0555)
Dependent Variable (1-lag)	-0.603***	-0.603***	-0.603***	-0.603***	-0.603***	-0.603***
	(0.0393)	(0.0393)	(0.0393)	(0.0393)	(0.0393)	(0.0393)
Dependent Variable (2-lag)	-0.442***	-0.442***	-0.442***	-0.442***	-0.442***	-0.442***
	(0.0708)	(0.0708)	(0.0708)	(0.0708)	(0.0708)	(0.0708)
Dependent Variable (3-lag)	-0.252***	-0.252***	-0.252***	-0.252***	-0.252***	-0.252***
	(0.0452)	(0.0452)	(0.0452)	(0.0452)	(0.0452)	(0.0452)
Expected Inflation (Level)	0.0957***	0.0951***	0.0956***	0.0953***	0.0954***	0.0953***
	(0.0332)	(0.0333)	(0.0333)	(0.0334)	(0.0333)	(0.0333)
Better Credit	-0.227	-0.225	-0.227	-0.228	-0.229	-0.231
	(0.380)	(0.381)	(0.380)	(0.380)	(0.380)	(0.380)
Better Finance	-0.297	-0.298	-0.290	-0.292	-0.287	-0.290
	(0.339)	(0.340)	(0.340)	(0.341)	(0.341)	(0.341)
Better Expected Finance	-0.882**	-0.880**	-0.873**	-0.871**	-0.870**	-0.874**
	(0.362)	(0.363)	(0.361)	(0.363)	(0.363)	(0.363)
Aggregate Uncertainty	-0.0218	-0.00665	-0.0271	-0.00128	-1.319	-3.564
	(0.0144)	(0.00466)	(0.0208)	(0.00146)	(1.060)	(2.345)
Constant	0.157	0.0282	0.187	-0.0956	0.850	3.083
	(0.880)	(0.873)	(0.915)	(0.880)	(1.204)	(2.320)
Controls	✓	✓	✓	✓	✓	✓
Joint IEU Effects	16.16	16.24	16.11	16.01	16.12	16.20
R-Squared	0.298	0.298	0.298	0.298	0.298	0.298
χ^2/F	53.98	54.03	53.87	54.23	54.01	54.11
Observations	39809	39809	39809	39809	39809	39809

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Controls included in the table: homeownership, female, race (whether the respondent is Caucasian), employment status, location (state), expected change in income, and current income indicators (less than \$30K, from \$30K to \$75K, from \$75K to \$200K, or \$200K or above). The dependent variable is the expected change in consumers' expenditure (in percentage). *Joint IEU Effects* denotes a joint test on the null hypothesis that the sum of the lag effects of inflation expectation uncertainty equals zero. Source: Author's calculations.

dependent variable of Eqn. (2) being expected, not contemporaneous, consumption.[11]

To align our results in this section with the previous exposition of real consumption in Eqn. (1), we also consider how the inclusion of various measures of aggregate uncertainty

[11]To examine whether the use of a fixed-effects panel in a dynamic regression setting gives rise to potential endogeneity concerns due to serial correlation in expected consumption growth, we also consider the Arellano-Bond GMM dynamic panel procedure instead of the benchmark fixed-effects panel and present these results in Table A.6 in the Appendix. Overall, we find that the additional results using the alternative GMM procedure are generally consistent with our benchmark fixed-effects panel regression for expected consumption.

impacts our expected consumption result. The first measure is the CBOE Volatility Index (*VIX*). The second and third measures are the CBOE Crude Oil ETF Volatility Index (*Oil*) and the CBOE Gold Volatility Index (*Gold*). The fourth measure is the Baker et al. (2016)'s Economic Policy Uncertainty Index for the U.S. (*EPU*). The fifth measure is the Jurado et al. (2015) macroeconomic uncertainty index at the three-month (*JLN-3*) and twelve-month (*JLN-12*) horizons. The results in Table 4, where the aggregate uncertainty measures in Columns (1)-(6) are *VIX, Oil, Gold, EPU, JLN-3,* and *JLN-12*, respectively, remain qualitatively consistent with the results in Table 3. Overall, the results in Table 4 suggest that the effects of IEU_{it} on expected consumption change are not driven by other channels of uncertainty that the literature has carefully documented.

2.2.3. Sensitivity Analyses

We consider five sensitivity analyses on the main results presented in this section for both real and expected consumption. First, we exclude data from the COVID-19 era. One notable feature of the data in our sample, as shown in Fig. 1, is the elevated inflation expectation uncertainty during the COVID-19 pandemic era that began in early 2020. Second, we exclude first-time survey respondents, given the tenure effects documented in the literature (Kim and Binder, 2023). Third, we consider an alternative measure of inflation expectation uncertainty, IEU_{it}: consumer i's inflation expectations' interquartile range in month t. Fourth, we control for aggregate inflation to examine the extent to which the effects of inflation expectation uncertainty documented simply reflect *actual* inflation movements. Fifth, we control for a measure of inflation tail risks to check if our measure of inflation expectation uncertainty broadly captures the inflation disaster risks documented in the literature (Ryngaert, 2022). Our real and expected consumption results are robust to these sensitivity analyses. For brevity, we relegate the details of these exercises in the Appendix.

2.3. Discussion

Our results indicate that an increase in inflation expectation uncertainty is related to a decrease in real and expected consumption. Given that real consumption captures whether the survey respondent has recently made a large purchase, and expected consumption is the percentage change in consumption expenditure in the next twelve months, this result implies that inflation expectation uncertainty not only induces a decrease in consumption in the current period but also has a negative and significant impact on expected consumption in the future.

This result, along with its consistency across different sensitivity analyses, suggests that respondents' individual financial and demographic characteristics do not drive the documented negative effect of inflation expectation uncertainty. Such negative results are not artifacts of aggregate uncertainty (Bloom, 2009; Jurado et al., 2015), the tenure effects of respondents (Kim and Binder, 2023), or inflation tail risks (Ryngaert, 2022). Indeed, these results complement the recent literature on general expectation uncertainty (e.g., Branch and McGough, 2009; Bachmann et al., 2015; Coibion and Gorodnichenko, 2015; De Grauwe and Macchiarelli, 2015, among others) and the effects of inflation expectation (e.g., Coibion et al., 2020, among others), as well as another broad literature on the implications of uncertainty on macroeconomic aggregates (e.g., Jurado et al., 2015 and Ludvigson et al., 2021, among others).

3. The Model

In Section 2, we have documented that both real and expected consumption respond negatively to an increase in inflation expectation uncertainty. The continued significance of inflation expectation uncertainty effects on consumption, even after controlling for common sources of aggregate uncertainty, suggests an economic environment in which agents exhibit uncertainty about next-period inflation.

To provide more structure to understand the negative consumption response documented in Section 2, we embed such an uncertainty channel in an otherwise standard New-Keynesian model in the spirit of Iacoviello (2005), in which the household purchases real estate, consumes goods, and works. Following Gabaix (2020), the household's inflation expectation is discounted because it might not be fully aware of the state of the economy. Unlike previous literature, the degree of cognitive discounting is time-varying and subject to an uncertainty shock that increases its distribution's dispersion.

3.1. The Household

The household chooses how much to consume C_{ht}, how much to work l_{ht} at the prevailing wage w_t, and how much to lend in real terms $-B_{ht}$. Following Iacoviello (2005), they also buy real estate H_{ht} at the price of q_t at time t. They maximize their lifetime discounted utility as follows

$$\max_{C_{ht}, H_{ht}, l_{ht}} \mathbb{E}_t \left[\sum_{s=0}^{\infty} \beta_h^s \left(\log C_{ht+s} + \log H_{ht+s} - \psi \frac{l_{ht+s}^{1+\xi}}{1+\xi} \right) \right], \tag{3}$$

subject to

$$C_{ht} + q_t \left(H_{ht} - H_{ht-1} \right) + \frac{R_{t-1} B_{ht-1}}{\pi_t} = B_{ht} + w_t l_{ht} + F_t, \tag{4}$$

in which q_t is the real house price, β_h is the household's subjective discount factor, F_t denotes the lump-sum profit from the retailers, and R_t is the gross nominal interest rate for the loan.

Motivated by the related empirical evidence, we assume that the household discounts future inflation expectation in the model.[12] More specifically, the household is subject to an exogenous cognitive discount factor m_t that alters the extent to which inflation expectation in period $t+1$ enters the household's decision in period t. Since the household does not fully understand the state of the economy, their expectation of next-period inflation $\mathbb{E}_t [\pi_{t+1}]$ is discounted by m_t. Operationally, m_t enters the consumer's first-order condition by replacing expected inflation $\mathbb{E}_t [\pi_{t+1}]$ with the cognitively discounted expected inflation $m_t \mathbb{E}_t [\pi_{t+1}]$. The interaction between m_t and $\mathbb{E}_t [\pi_{t+1}]$ allows the cognitive discount factor m_t to impact

[12] For example, De Groote and Verboven (2019) show that households significantly discount future benefits of new technology. Inoue et al. (2009) use micro evidence to show that households' inflation expectations do respond to inflation news, suggesting a higher subjective weight on near-term inflation expectations when information regarding future inflation is not readily available.

the distribution of $\mathbb{E}_t [\pi_{t+1}]$ and how the expectation term enters the household problem. We assume that only the inflation expectation in the model is subject to cognitive discounting.

In our empirical analysis, we have documented that inflation expectation uncertainty has a significant effect on consumption, even after controlling for other sources of aggregate uncertainty. Our assumption that cognitive discounting applies to inflation expectations is partly motivated by this empirical result. In addition, doing so allows us to cleanly isolate the effects of inflation expectation uncertainty in the model, effectively mirroring the empirical analysis, which controls for other sources of uncertainty.

The household cognitive discount factor m_t is assumed to follow an AR(1) process as follows

$$\log m_t = (1 - \rho_M) \log m + \rho_M \log m_{t-1} + \sigma_{Mt} \varepsilon_t^M \tag{5}$$

$$\log \sigma_{Mt} = (1 - \rho_{SM}) \log \sigma_M + \rho_{SM} \log \sigma_{Mt-1} + \nu_t, \tag{6}$$

in which m is the steady-state cognitive discount factor, and σ_{Mt} is the stochastic volatility component associated with the m_t process. The exogenous shock ε_t^M is i.i.d. and $\varepsilon_t^M \sim N(0, 1)$. ν_t is also i.i.d. and normally distributed with mean zero. ρ_M is the persistence parameter for the m_t process, and ρ_{SM} is the persistence parameter of its stochastic volatility component. As noted by the related literature (Fernández-Villaverde et al., 2011), an unexpected shock ν_t to the stochastic volatility component σ_{Mt} impacts other endogenous variables in the model via the higher-order terms, which can lead to a change in m_t, depending on the direction of the shock.[13]

[13] Hirose et al. (2024) estimate Gabaix-style cognitive discounting parameters in a full DSGE with Bayesian methods. They find that models with bounded rationality show better fit to the U.S. data than its rational expectations counterpart.

3.2. The Entrepreneur

The entrepreneur owns a firm that produces intermediate goods using the following technology

$$Y_t = A_t H_{et-1}^{\alpha} L_{et}^{1-\alpha}, \tag{7}$$

in which α is the share of commercial real estate in production, H_{et-1} is the stock held at the end of period $t-1$, and L_{et} represents the total labour demand in the economy. The total factor productivity (TFP) A_t follows the exogenous AR(1) process $\log A_t = \rho_A \log A_{t-1} + \varepsilon_t^A$, where ρ_A denotes the persistence of the TFP process and $\varepsilon_t^A \sim N\left(0, \sigma_A^2\right)$. The entrepreneur maximizes their lifetime utility from consumption C_{et} as follows

$$\max_{C_{et}, H_{et}, L_{et}, B_{et}} \mathbb{E}_t \left[\sum_{s=0}^{\infty} \beta_e^s \log C_{et+s} \right], \tag{8}$$

subject to

$$C_{et} + q_t \left(H_{et} - H_{et-1}\right) + w_t L_{et} + \frac{R_{t-1} B_{et-1}}{\pi_t} = \frac{Y_t}{X_t} + B_{et}, \tag{9}$$

where X_t is the markup in period t. The entrepreneur must use their real estate holding as collateral when borrowing; that is,

$$B_{et} \leq \Xi_e \mathbb{E}_t \left(q_{t+1} H_{et} \frac{\pi_{t+1}}{R_t} \right), \tag{10}$$

in which Ξ_e is the entrepreneur's loan-to-value ratio. Our motivation for including this collateral constraint follows from the literature that highlights the role of financial frictions (see, e.g., Kiyotaki and Moore, 1997 and Bernanke et al., 1999) and the housing market (see, e.g., Iacoviello, 2005). Indeed, in Section 5.3, we find that tightening collateral constraint (via a lower Ξ_e) alters the responses of key macroeconomic aggregates to shocks to inflation expectation uncertainty.

3.3. The Retailers

Following Bernanke et al. (1999), the retailers purchase the intermediate goods

$$Y_t = \left(\int_0^1 Y_t\left(j\right)^{(\theta-1)/\theta} dj \right)^{\theta/(\theta-1)}, \tag{11}$$

where $Y_t\left(j\right)$ denotes the output of retailer j, which is produced by the firm at a wholesale price, competitively, and distributed in a monopolistically competitive market. The demand for each retailer is $Y_t\left(j\right) = \left(P_t\left(j\right)/P_t\right)^{-\theta} Y_t$. In each period, every retailer is subject to a probability γ of not being able to change their price, and thus γ is a measure of price rigidity. The optimal reset price $P_t^*\left(j\right)$ of retailer j solves

$$\sum_{s=0}^{\infty} \gamma^s \mathbb{E}_t \left[\Lambda_{ts} \left(\frac{P_t^*\left(j\right)}{P_{t+s}} - \frac{X}{X_{t+s}} \right) \left(\frac{P_t^*\left(j\right)}{P_{t+s}} \right)^{-\theta} Y_{t+s} \right] = 0, \tag{12}$$

in which $\Lambda_{ts} = \beta_h^s \mathbb{E}_t\left[C_{ht}/C_{ht+s}\right]$ is the stochastic discount factor from the household problem and $X = \theta/\left(\theta-1\right)$ is the flexible-price steady-state markup.

3.4. Home Construction

In every period, constructors maximize their profit, subject to a quadratic housing adjustment cost

$$\max_{I_t} \mathbb{E}_t \left[q_t I_t - I_t - \frac{\chi}{2} \left(\frac{I_t}{H_t} \right)^2 H_t \right], \tag{13}$$

and $I_t = H_t - H_{t-1}$, where $H_t = H_{et} + H_{ht}$ and I_t denote the total housing stock and total housing investment at time t, respectively. The adjustment cost $\frac{\chi}{2}\left(\frac{I_t}{H_t}\right)^2 H_t$ is assumed to be not resource-consuming.

3.5. *Monetary Policy and Market Clearing Conditions*

To close the model, monetary policy follows a standard Taylor rule

$$R_t = R_{t-1}^{\rho_R} \left(R \left(\frac{Y_t}{Y} \right)^{\phi_Y} \left(\frac{\pi_t}{\pi} \right)^{\phi_\pi} \right)^{1-\rho_R} \exp\left(\varepsilon_t^R \right), \tag{14}$$

where ρ_R denotes the degree of interest-rate smoothing, ε_t^R is i.i.d., and $\varepsilon_t^R \sim N\left(0, \sigma_R^2\right)$. Y and π denote the steady-state output and the target inflation rate, respectively. The markets for consumption goods, labour, and housing must clear; that is, $Y_t = C_{et} + C_{ht} + I_t$, $L_{et} = l_{ht}$, and $H_t = H_{et} + H_{ht}$. Lastly, the loan market must clear; that is, $B_{ht} + B_{et} = 0$.

4. Parameters and Model Estimation

We calibrate several parameters for the model and estimate the remaining parameters using the simulated method of moments (SMM) in the spirit of Fernández-Villaverde et al. (2015). The impetus behind such estimation is to help discipline our model to check its qualitative consistency with the empirical results in Section 2.

4.1. *Parameters Set to External Targets*

Some parameters are standard in the New-Keynesian literature and are set based on existing studies. The household's subjective discount factor, β_h, is 0.99, reflecting an annualized steady-state interest rate of around 4%. Following Iacoviello (2005), the model's structure for the entrepreneur yields a subjective discount factor of 0.98, β_e. Similarly, the labour disutility parameter ψ governing the leisure weight relative to consumption in household preferences is 1, and the inverse of the Frisch elasticity of labour supply ξ is 0. Our choice of $\xi = 0$ follows the literature on indivisible labour supply (Hansen, 1985; Rogerson, 1988). The share of commercial real estate in production, α, is set to 0.33, in line with standard capital-share values in the literature.

Turning to the parameters on production and nominal rigidity, we set γ, the probability

Table 5: Parameters

(a) Calibrated Parameters

Parameter	Value
Preferences	
β_h	0.99
β_e	0.98
ψ	1
ξ	0
Non-preference Parameters	
α	0.33
γ	0.75
θ	5
Ξ_e	0.89
χ	1
π	$1.02^{1/4}$

(b) Estimated Parameters

Parameter	Value
Model Parameters	
m	0.752
ρ_M	0.922
ρ_A	0.899
ρ_{SM}	0.211
ρ_R	0.837
ϕ_Y	0.119
ϕ_π	1.499
Standard Deviations of Shocks	
σ_A	0.0031
σ_R	0.0005
σ_v	0.0080

Source: Author's calculations.

that retailers are stuck with the current-period price, to 0.75, which implies an average time between price changes of around 4 quarters (Dennis, 2006; Schorfheide, 2008). The elasticity of substitution between the differentiated goods, θ, is 5, which falls within the commonly used values in the literature. For the loan-to-value ratio, we follow Iacoviello (2005) and set $\Xi_e = 0.89$. The weight for the housing adjustment cost χ is 1. The inflation target π equals $1.02^{1/4}$, reflecting a long-run target inflation rate of around 2%. Panel (a) of Table 5 summarizes all parameters set to external targets.

4.2. Estimation and Solution Method

We estimate the remaining parameters m, ρ_M, ρ_{SM}, ρ_A, the parameters governing the Taylor rule, along with the standard deviations of the shocks using the simulated method of moments in the spirit of Ruge-Murcia (2012) and Fernández-Villaverde et al. (2015). In particular, we match the model to multiple moments generated from the following series: real consumption expenditure growth, real output growth, inflation, and inflation expectation. All quarterly data are from FRED (Federal Reserve Bank of St. Louis) and from 1984:Q1 to

Table 6: Model and Data Moments

(a) Moments Related to Prices and Expected Inflation

Moments	Model ($\times 10^{-3}$)	Data ($\times 10^{-3}$)
$\pi_t^M \pi_t$	0.666	0.551
$\pi_t^M \pi_{t-1}$	0.604	0.504
$\pi_t^M \pi_{t-2}$	0.507	0.461
$\pi_t^M \pi_{t-3}$	0.435	0.421
$\pi_t^M \pi_{t-4}$	0.311	0.384
π_t^2	0.008	0.007
$\pi_t \pi_{t-1}$	0.007	0.006
$\pi_t \pi_{t-2}$	0.006	0.006

(b) Moments Related to Output

Moments	Model	Data
$\tilde{y}_t^2$	0.340	0.314
$\tilde{y}_t \tilde{c}_t$	0.199	0.247
$\tilde{y}_t \tilde{c}_{t-1}$	0.115	0.104
$\tilde{y}_t \pi_t$	0.277	0.095
$\tilde{y}_t \tilde{y}_{t-1}$	0.112	0.044
$\tilde{y}_t \tilde{y}_{t-2}$	0.051	0.002

Notes: $\tilde{y}_t$ and $\tilde{c}_t$ denote the growth rate of output and consumption, respectively. π_t^M and π_t denote expected inflation and inflation, respectively. Data are from FRED (Federal Reserve Bank of St. Louis) and are from 1984:Q1 to 2019:Q4. Additional details on the transformations of the series used are in the appendix. Source: Author's calculations.

2019:Q4. This estimation sample is deliberately longer than, and does not coincide with, the 2013:M6–2023:M9 survey sample used in Section 2 or the 2013–2023 aggregate sample used in Section 6, since the aggregate series used to construct the moments are available over a much longer horizon. Since the model contains a stochastic volatility component, following Fernández-Villaverde et al. (2011) and Born and Pfeifer (2014), we solve and estimate it using third-order perturbation. Further details on the series and the moments used in the estimation are in the Appendix. Panel (b) of Table 5 provides the estimates of m, ρ_M, ρ_{SM}, ρ_A, the Taylor rule parameters, and the standard deviations of the shocks used in the model.[14]

Our estimation yields a persistence parameter, ρ_M, of 0.922 for the cognitive discount shock process. The steady-state cognitive discount factor m is estimated to be 0.752. This value implies that the information the agents receive today has a half-life of around two and a half quarters. This number is in the ballpark of other values found in the literature. For example, Gabaix (2020) uses a value of 0.85, implying an information half-life of around a year, whereas Fuhrer and Rudebusch (2004) estimate a value (specifically, the weight on expected future output) of 0.65 and Benchimol et al. (2025) find such a value to be between

[14] In the appendix, we vary selected parameter values to check on the model's implications. Indeed, as noted by Kozlowski et al. (2020), tail events can persistently shift beliefs about uncertainty.

0.76 and 0.98. Finally, the Taylor rule parameter estimates are broadly consistent with the related literature (Taylor, 1999, Judd and Rudebusch, 1998, Clarida et al., 2000, Orphanides, 2004, and Coibion and Gorodnichenko, 2012).

To examine how the model aligns with the data, Table 6 presents the model's selected moments and the data's moments. Overall, the model's moments are generally consistent with their empirical counterparts. Specifically, apart from three simulated moments related to output, the model matches the data reasonably well across moments for prices, consumption, and output. For brevity, we leave additional details on the construction and transformations of these moments in the Appendix.

5. Discussion

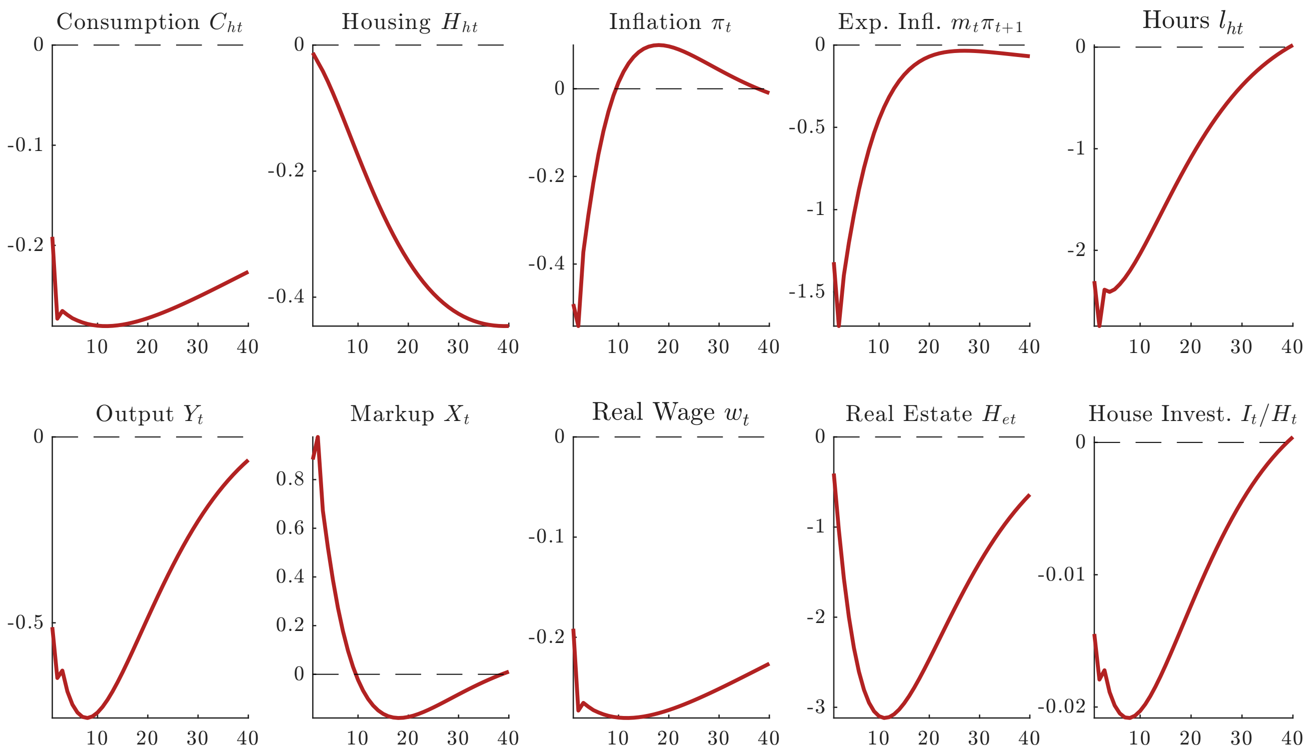
We next discuss the model responses to a positive shock to the inflation expectation uncertainty. A positive shock to the stochastic volatility component leads to a contraction in consumption and other macroeconomic aggregates.

5.1. Model Responses to Inflation Expectation Uncertainty Shocks

We initiate a positive one-standard-deviation shock to ν_t, the innovation term of the stochastic volatility component of the cognitive discount process m_t, and calculate the responses of household consumption C_{ht}, housing consumption H_{ht}, inflation π_t, expected inflation $\mathbb{E}_t\left[m_t \pi_{t+1}\right]$, labour hours l_{ht}, output Y_t, markup X_t, real wage w_t, commercial real estate holdings H_{et}, and housing investment I_t. We consider this shock to ν_t to be the inflation expectation uncertainty shock as it amplifies the stochastic volatility component of the m_t process. Since an unexpected increase in ν_t widens the distribution of a shock to m_t, it impacts aggregate demand via the extent to which the household decides on the next-period consumption.

Fig. 2 plots the impulse responses of selected variables to a positive one-standard-deviation shock to ν_t. The unit for inflation responses is percentage points. Since housing investment

Figure 2: Responses to Inflation Expectation Uncertainty Shock



Notes: Except for inflation, expected inflation, and housing investment, all other responses are in terms of percentage deviations from the corresponding steady states. The unit for inflation responses is percentage points. Since housing investment I_t is zero in the steady state, we normalize the response relative to the total housing stock at time t. The unit of housing investment I_t is the deviation from the steady state relative to the housing stock. Source: Author's calculations.

I_t is zero in the steady state, we divide its response by the total housing stock at time t, and thus, the unit of housing investment I_t is the deviation from its steady-state level relative to the housing stock. All other responses are percentage deviations from the corresponding steady states. [15]

Fig. 2 shows that household consumption falls relative to its steady-state level following a positive one-standard-deviation shock to inflation expectation uncertainty. This decline is accompanied by a persistent decrease in housing investment, housing consumption, wages, and output, and a rise in markup. The troughs of these responses, except for the consumption of housing services H_{ht}, are within the first eight to ten quarters after the initial shock.

[15] For brevity, we leave the remaining responses in the Appendix.

5.2. *Intuition*

To understand the extent to which an increase in the mean-preserving spread of the cognitive discount factor m_t can impact consumption in Section 5.1, we consider an expression for expected consumption growth

$$\mathbb{E}_t\left(\frac{C_{ht+1}-C_{ht}}{C_{ht}}\right)=\frac{\frac{1}{2m}\mathbb{V}_{ct}+\frac{1}{m}\mathbb{V}_{mt}+a\left(m_t,\pi_t\right)+\frac{1}{m}\mathbb{V}_{\pi t}+\frac{1}{m}-\frac{1}{\beta_h R}}{b\left(m_t,\pi_t\right)},\tag{15}$$

which is a second-order approximation of the Euler equation around the point $C_{ht+1} = C_{ht}, m_t = m$, and $\pi_t = 1$. $\mathbb{V}_{ct}$, $\mathbb{V}_{mt}$ and $\mathbb{V}_{\pi t}$ denote the mean-preserving volatility components of consumption, the cognitive discount factor, and inflation, and $a\left(m_t,\pi_t\right)$ and $b\left(m_t,\pi_t\right)$ are expressions of first-moment terms of m_t and π_t.[16] A rise in the mean-preserving spread of the cognitive discount factor $\mathbb{V}_{mt}$ via a positive shock to the stochastic volatility component of the cognitive discount factor increases the expected consumption growth, leading to a fall in the current-period household consumption due to precautionary saving motives (Basu and Bundick, 2017).

Two additional channels amplify the decline in consumption. The first is through increased markup, and the second is through inflation expectation distortion. We will explore their implications in the following subsections.[17]

5.2.1. *The Markup Channel*

As prices are not perfectly flexible, a surge in markup induces firms to produce less output, requiring fewer labour hours and less commercial real estate. As shown in Fig. 2, labour hours and commercial real estate decline as the markup X_t increases following a positive shock to the stochastic volatility component σ_{Mt} of the cognitive discount factor m_t. Such falls

[16] Here, $a\left(m_t,\pi_t\right) = \left(m_t-m\right)\left(\frac{1}{m^2}\right)\left(\pi_t-1\right)$ and $b\left(m_t,\pi_t\right) = \frac{2}{m}-\frac{\left(m_t-m\right)}{m^2}-\frac{\pi_t}{m}$. While a third-order approximation is needed to observe the effect of a shock to ν_t on $\mathbb{V}_{mt}$, we use a second-order approximation for expositional convenience since our focus here is on the effects of $\mathbb{V}_{mt}$ on expected consumption growth.

[17] Relatedly, Ambrocio (2026) uses Euro area household survey uncertainty and finds markup increases and precautionary savings consistent with the channels considered here.

in demand for commercial real estate and labour hours put downward pressure on household consumption of goods and housing services.

Following the positive shock to ν_t, inflation falls but gradually reverts to its steady-state level thereafter. As the central bank follows the Taylor rule, the nominal interest rate initially falls, but not enough to offset the rise in the markup, leading to a sustained decline in consumption. This markup channel is similar to that found in the related literature on the effects of volatility shocks (e.g., Fernández-Villaverde et al., 2011, Fernández-Villaverde et al., 2015, and, more recently, Born and Pfeifer, 2021). Moreover, because retailers cannot change their prices with probability γ, prices might not fully adjust to the downward pressure on aggregate demand. Such nominal rigidity amplifies the negative effects of inflation expectation uncertainty on consumption.

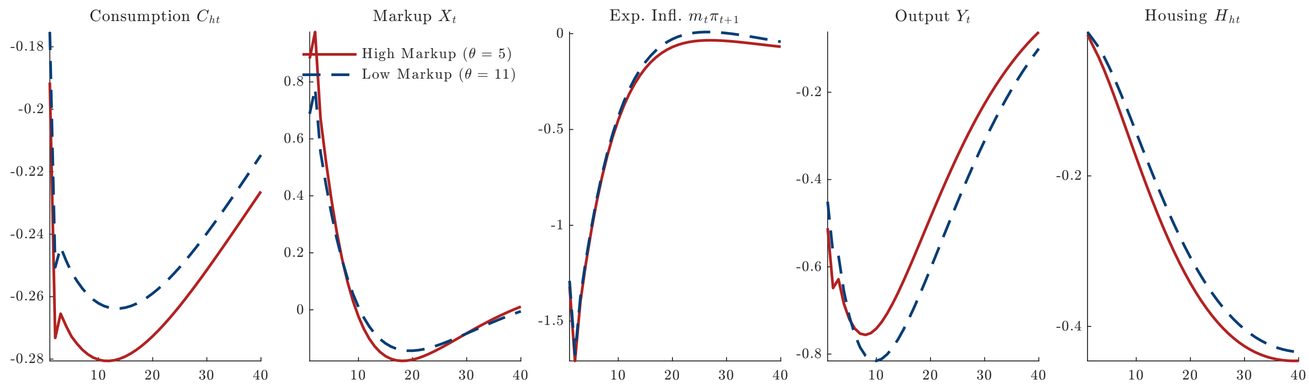
To further inspect the markup channel in the model, Fig. 3a plots the responses of consumption, markup, expected inflation, output, and housing consumption to the same shock used in Fig. 2, but with $\theta = 11$ (the dashed lines) in addition to the case in which $\theta = 5$ in the baseline (the solid lines). A value of $\theta = 11$ implies a steady-state markup of roughly 10%, smaller than the steady-state markup of around 25% used in the baseline case. The impetus of this exercise is to examine how the model responds when the markup channel is subdued.

Fig. 3a shows that a fall in the steady-state markup ($\theta = 11$, the dashed lines) induces a smaller increase in the markup response (X_t) than under the baseline model ($\theta = 5$, the solid lines). This smaller increase implies lower pressure on the firm to reduce hiring and demand less commercial real estate, resulting in less negative consumption and output responses on impact, as observed in Fig. 3a.[18] Specifically, under the lower markup case ($\theta = 11$, the dashed lines), both output and consumption fall less on impact and over the subsequent quarters than under the higher markup case ($\theta = 5$, the solid lines). Housing demand also falls less under the lower markup case than the benchmark one.

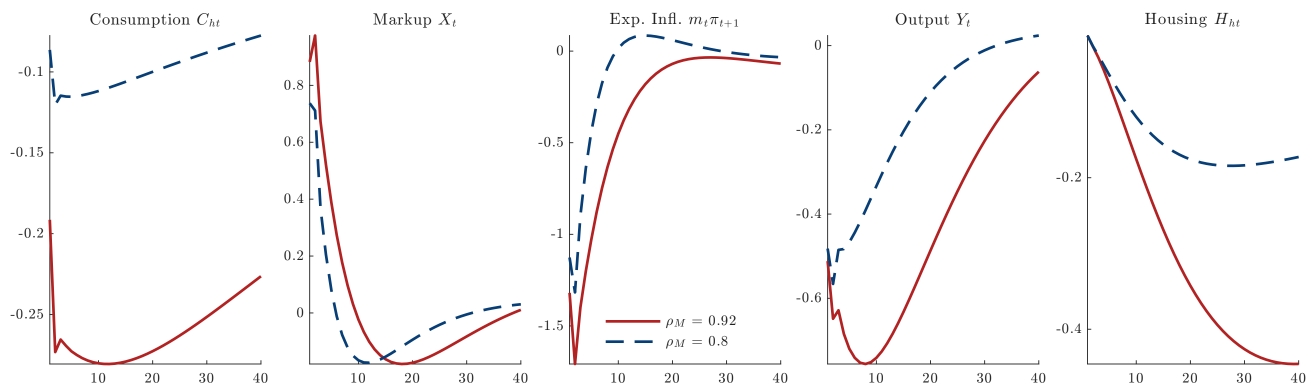
[18]Section D in the Appendix reports additional results that alter our benchmark model in several ways to check their robustness and our interpretations of the findings.

Figure 3: Inspecting the Model's Mechanism

(a) The Markup Channel



(b) The Expectation Channel



Notes: Except for inflation, all other responses are in terms of percentage deviations from the corresponding steady states. The unit for inflation responses is percentage points. Source: Author's calculations.

5.2.2. The Expectation Channel

Following a positive shock to the stochastic volatility component of the cognitive discount factor m_t, the expected inflation is distorted, evidenced by the negative responses of $\mathbb{E}_t\left[m_t\pi_{t+1}\right]$ in Fig. 2. This fall in inflation expectation increases the expected real interest rate. It prompts households to save more, consistent with the overall decline in consumption. While the central bank responds to the decline in inflation by cutting interest rates, its efforts do not sufficiently offset the expectation channel, as it responds to the actual inflation rate rather than the distorted inflation expectation.

To understand the importance of the expectation channel in generating our result, Fig. 3b plots the responses of consumption, markup, expected inflation, output, and housing consumption to the same shock in Fig. 2, but with $\rho_M = 0.8$ (the dashed lines) in addition to the case in which $\rho_M = 0.92$ in the baseline (the solid lines). Mechanically, a smaller value of ρ_M implies dampened propagation of the inflation expectation uncertainty shock ν_t. Fig. 3b shows that consumption responses are subdued as the distorted inflation expectation reverts to its pre-shock level more quickly. This result is consistent with the amplification of the expectation channel we considered in Fig. 2.

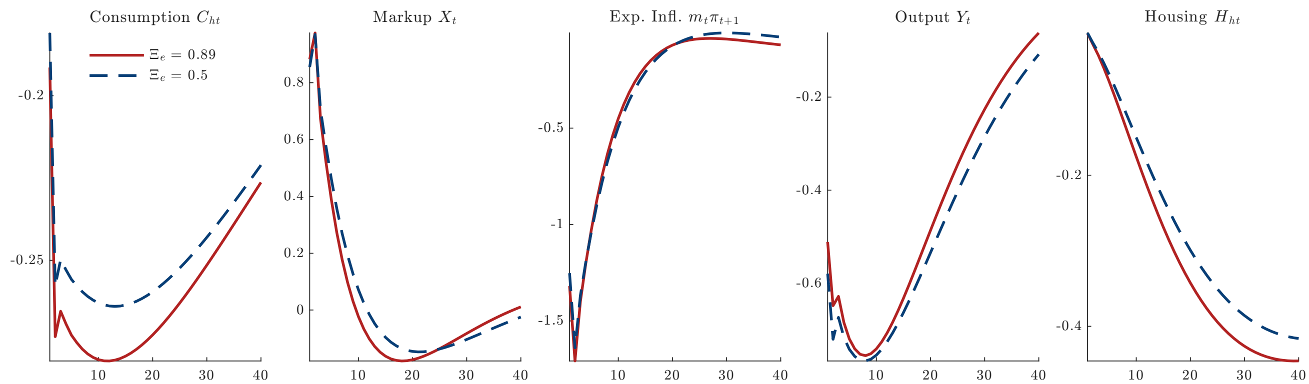
Overall, the combination of the markup and expectation channels amplifies the precautionary saving motives, allowing a positive stochastic volatility shock to m_t to induce the falls in consumption and other macroeconomic aggregates. More importantly, this model prediction is qualitatively consistent with the empirical evidence from the regression analysis in Section 2.2.

5.3. The Role of Financial Frictions

The collateral constraint limits the entrepreneur's borrowing to a fraction Ξ_e of the expected value of their commercial real estate holdings. Intuitively, when the IEU shock negatively affects housing demand and house prices, the value of the entrepreneur's collateral falls, limiting their ability to borrow (Iacoviello, 2005). In this section, we examine the quantitative implications of this channel by comparing the baseline model ($\Xi_e = 0.89$) with an economy in which the collateral constraint is tighter ($\Xi_e = 0.5$).

Fig. 4 plots the impulse responses of selected variables under both parameterizations. Under the tighter constraint, the output response to the IEU shock is more negative than under the baseline. This is because the entrepreneur operates with less financial slack and must cut production more aggressively when borrowing capacity contracts. At the same time, consumption and housing decline less under $\Xi_e = 0.5$ than under the baseline, as the smaller markup increase arising from the IEU shock (shown in Fig. 4) translates into weaker downward pressure on consumption and housing demand. Expected inflation responses are roughly comparable under $\Xi_e = 0.5$ and under the baseline, which suggests that the effects of the expectation channel are largely consistent across these two cases.

Figure 4: Responses to Inflation Expectation Uncertainty: The Role of Financial Frictions



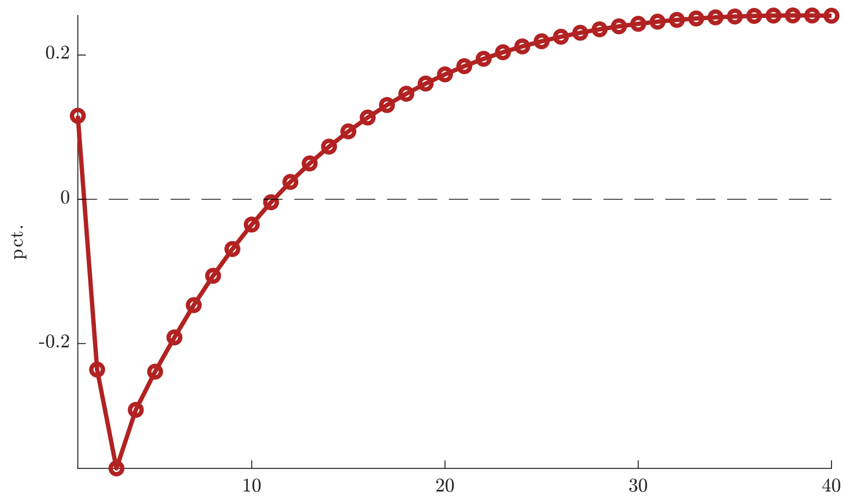
Notes: Except for inflation, all other responses are in terms of percentage deviations from the corresponding steady states. The unit for inflation responses is percentage points. Source: Author's calculations.

These results show that financial frictions shape the composition of the transmission mechanism rather than simply amplifying or dampening the aggregate response. A tighter collateral constraint redirects the effects of the IEU shock toward output, as constrained entrepreneurs absorb a larger share of the adjustment, while partially insulating consumption through the muted markup channel. In our empirical analysis, we find that the respondent's financial conditions (i.e., *Better Finance*) are statistically significant in explaining consumption (Table 1), consistent with the model prediction that the degree of financial frictions matters for how inflation expectation uncertainty transmits to the real economy.

6. Model Predictions, Survey Evidence, and Aggregate Consumption

Since our empirical analysis relies primarily on survey data, a natural question arises as to how the model, the survey data, and the aggregate data are connected. Here, we attempt to connect the three with a focus on how the model's implications can be interpreted in the confines of both survey and aggregate data. To that end, we first discuss the explicit connection between the measure of expected consumption in the model and that from the survey data. We then move on to the aggregate evidence and discuss how our measure of inflation expectation uncertainty aggregates up and how such a measure is commensurate with each of the components in the aggregate consumption baskets.

Figure 5: Responses of Expected Consumption Growth to IEU



Notes: The figure plots the response of expected consumption growth to a positive one-standard-deviation shock to inflation expectation uncertainty. Source: Author's calculations.

6.1. Expected Consumption in the Model and from the Survey Data

Fig. 5 plots the response of expected consumption growth $\mathbb{E}_t\left[(C_{ht+1} - C_{ht})/C_{ht}\right]$ to a positive one-standard-deviation shock to inflation expectation uncertainty. On impact, expected consumption growth rises, consistent with the precautionary saving mechanism in Eqn. (15). Specifically, the increase in the mean-preserving spread of the cognitive discount factor $\mathbb{V}_{mt}$ raises expected consumption growth, which in turn depresses current-period consumption as the household saves more. After the initial period, however, expected consumption growth turns negative and remains so for multiple quarters as the shock propagates through the markup and expectation channels discussed in Section 5.[19]

These responses align with the empirical results in Table 3. The positive contemporaneous coefficient on IEU in Column (1) aligns with the initial positive response of expected consumption growth in Fig. 5. The positive but imprecisely estimated contemporaneous coefficients on IEU in Column (2) onward are directionally consistent with the initial positive

[19]Our result on the response of expected consumption growth echoes that of Macaulay and Moberly (2026), who use German household survey data to show heterogeneity in expectation formation amplifies aggregate consumption responses by an order of magnitude.

response of expected consumption growth in the figure. At the same time, the largely negative and statistically significant lagged coefficients on IEU, which drive the overall joint effects reported at the bottom of Table 3, are consistent with the negative expected consumption growth responses in subsequent periods. In other words, the model predicts that expected consumption growth initially rises as households adjust their savings behavior, then falls as the adverse effects of inflation expectation uncertainty propagate through the economy. The survey data thus support both predictions.

We note that the model's expected consumption growth $\mathbb{E}_t\left[(C_{ht+1} - C_{ht})/C_{ht}\right]$ is a conditional expectation computed under the household's (cognitively discounted) information set, whereas the survey's dependent variable in Table 3 captures each respondent's subjective belief about how their total spending will change over the next twelve months. The two objects need not coincide, as the subjective report may also reflect sentiment, salience of recent price experiences, or other factors outside the model (see, e.g., D'Acunto and Weber, 2024). We therefore interpret the mapping between Fig. 5 and Table 3 as directional rather than quantitative.

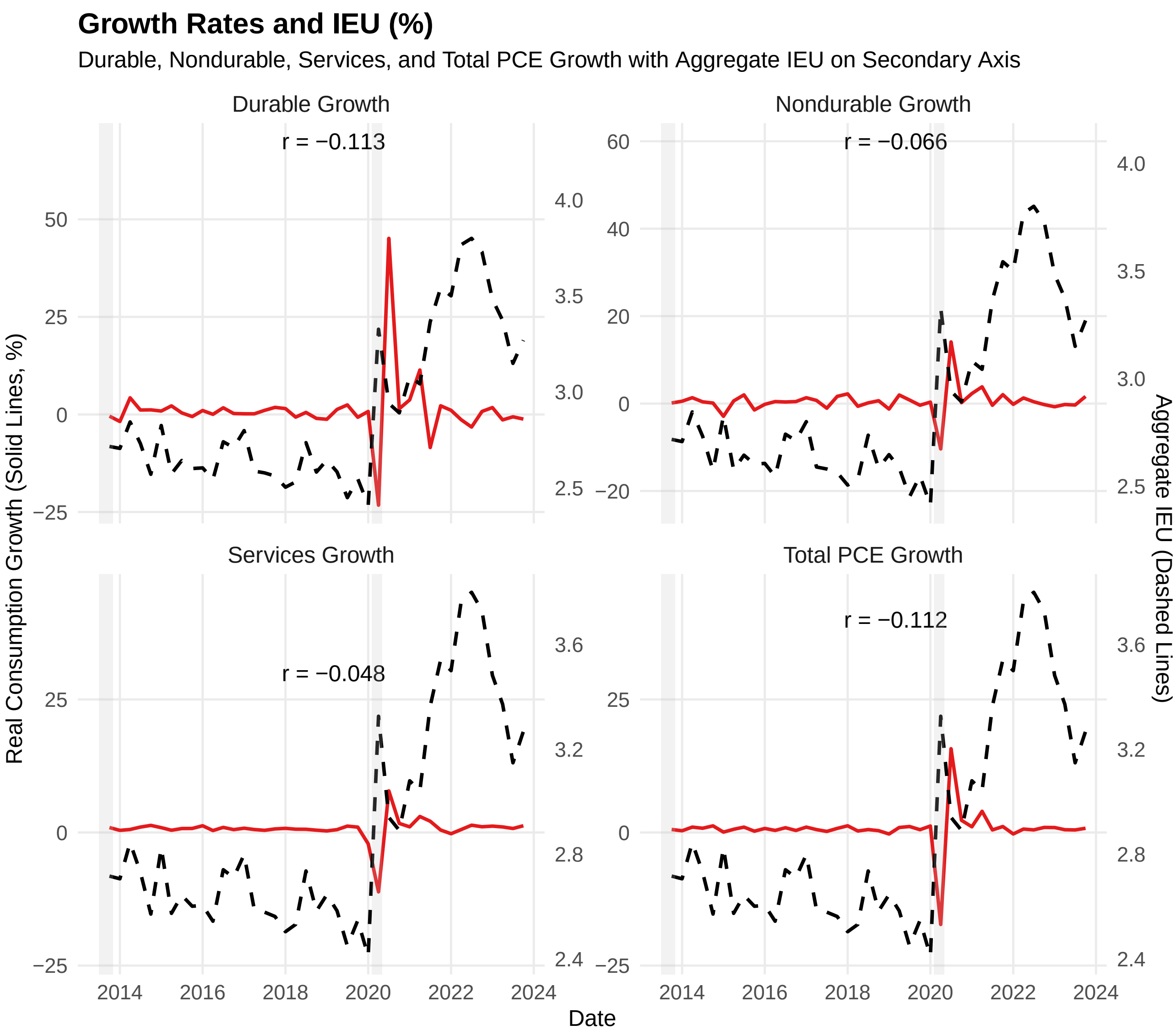
6.2. Aggregate IEU and Consumption Components

In the empirical section, we documented that both actual and expected consumption measures are negatively related to an unexpected increase in inflation expectation uncertainty (IEU) at the survey respondent level. A remaining concern is that the consumption variable used in the survey captures only a relatively narrow category of spending, namely, large durable purchases, rather than total consumption. We address this concern by examining the extent to which durable goods consumption co-moves with broader consumption aggregates and responds similarly to changes in IEU.

Panel (a) of Table 7 presents descriptive statistics for the quarterly growth rates of total personal consumption expenditure (PCE), durables, nondurables, and services over our sample period, along with their correlations with aggregate IEU and total PCE.[20] Durable

[20]Aggregate IEU is the average of the individual inflation expectation uncertainty measure across all survey

Figure 6: Real Aggregate Consumption and Average Inflation Expectation Uncertainty



Notes: The figure plots the aggregate consumption growth (total PCE, durables, nondurables, and services) against the average inflation expectation uncertainty across all survey respondents. Consumption growth and its components are plotted as solid lines, with the aggregate IEU represented by the dashed line. All aggregate data are from FRED (Federal Reserve Bank of St. Louis). Durable goods are based on the series *PCEDG*. Nondurable goods are based on the series *PCEND*. Service goods are based on the series *PCESV*. Total PCE is based on the series *PCE*. All consumption data are deflated using the series *GDPDEF*. Source: FRED and the author's calculations.

goods consumption is roughly twice as volatile as total PCE (8.40% versus 3.77%) and is the consumption category that exhibits the most negative correlation with aggregate IEU (-0.113). Nondurables and services, which largely comprise of necessities such as food, rent, and util-

respondents. All consumption data are from FRED (Federal Reserve Bank of St. Louis) and are seasonally adjusted.

ities, exhibit weaker negative correlations with IEU. Such a dichotomy aligns with the view that durable goods purchases represent the postponable, discretionary margin of consumption where uncertainty is most likely to affect household decisions. At the same time, all three components are strongly correlated with total PCE ($r > 0.90$). This confirms that the consumption components share common cyclical dynamics.

Turning to Panel (b) of Table 7, we compare the average growth rates of each consumption category across NBER recessions and expansions. All categories contract during recessions, but durables decline the most: -11.84% on average, compared with roughly -5% for nondurables and services and -8.36% for total PCE. Such cyclical patterns are consistent with the related literature, which documents that durable goods are the most sensitive margin of consumption over the business cycle (see, e.g., Mankiw, 1985, Barsky et al., 2007, and Erceg and Levin, 2006). Since durable purchases are lumpy, often financed, and easily postponed, they respond most strongly to changes in economic conditions. While the recession in our sample corresponds to the COVID-19 contraction, the fall in durables is consistent with the evidence from earlier recessions documented in the literature. If inflation expectation uncertainty depresses consumption across categories, it should be most pronounced in durables, which is precisely what our survey regressions document.

Fig. 6 plots the quarterly growth rate of real consumption (total PCE) and its components (durables, nondurables, and services) alongside aggregate IEU on the secondary axis. All panels show that episodes of elevated IEU coincide with weaker consumption growth, and this association is visually more pronounced for durables than for total PCE and the remaining two components, in the sense that durables swing by substantially more around high-IEU episodes.[21] The correlations between IEU and consumption growth are similar in magnitude, at -0.113 for durables and -0.112 for total PCE, confirming that the negative association documented in our survey regressions is also present in the aggregate data. While these aggregate correlations appear somewhat weaker than those at the survey level, given the limited time dimension and the noise in quarterly aggregates, they nonetheless support

[21] Kim and Wong (2025) find significant quantitative differences in adjustment costs among durable goods categories, which may help explain why durables can be more sensitive to uncertainty than other categories.

Table 7: Aggregate Consumption Components vs. IEU (2013-2023)

Panel (a): Mean, Standard Deviation, and Correlations

Variable	Mean Growth (%)	SD (%)	Corr with IEU	Corr with Total PCE
Total PCE	0.700	3.770	-0.112	1.000
Durables	1.083	8.405	-0.113	0.906
Nondurables	0.527	3.004	-0.066	0.930
Services	0.683	2.301	-0.048	0.951

Panel (b): Average Growth Rates: Recession vs. Expansion

Period	Total PCE (%)	Durables (%)	Nondurables (%)	Services (%)
Expansion	1.164	1.746	0.817	0.98
Recession	-8.358	-11.843	-5.128	-5.115

Notes: Growth rates are quarter-over-quarter percentages computed from monthly PCE data aggregated to quarterly totals. IEU (Inflation Expectation Uncertainty) is averaged across all survey respondents. Correlations are computed using growth rates from 2013 to 2023. All aggregate data are from FRED (Federal Reserve Bank of St. Louis). Durable goods are based on the series *PCEDG*. Nondurable goods are based on the series *PCEND*. Service goods are based on the series *PCESV*. Total PCE is based on the series *PCE*. All consumption data are deflated using the series *GDPDEF*. Source: Author's calculations.

the model's negative responses at both the aggregate and micro levels.

7. Conclusion

This paper examines how uncertainty about inflation expectations affects consumption. Using survey data on expectations and consumption, we document that a rise in inflation expectation uncertainty is associated with a fall in both real and expected consumption spending. This result is robust across various sources of aggregate uncertainty, aggregate price movements, and household demographic and financial characteristics. The robustness of our empirical results when controlling for various sources of aggregate uncertainty suggests that inflation expectation uncertainty is an essential factor in understanding changes in consumption spending.

To understand the potential mechanism behind these empirical results, we consider an economic environment in which agents are uncertain about their expectations of next-period inflation. Specifically, we study a dynamic New-Keynesian model in which the household cognitively discounts its future inflation expectation. We find that an unexpected increase in uncertainty about the household's degree of cognitive discounting can generate a negative consumption response that is qualitatively consistent with the empirical analysis. Such consistency in consumption responses across the empirical analysis and the model provides a potential mechanism for understanding the empirical result. In addition to precautionary saving motives, increases in markups and expected real interest rates arising from inflation expectation distortions amplify the fall in consumption. Given the role of inflation expectation uncertainty documented in this paper and the importance of these expectations in shaping monetary policy, future research could explore the interplay between these two channels.

Supplementary Material

Supplementary material is available on the OUP website. It comprises the online appendix, data, and replication files.

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Conflict of Interest

The author declares that there are no conflicts of interest.

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